Name ____ Date ___________ Period _____

Parent Functions II: x^3 and $\sqrt[3]{x}$

Algebra II Honors | Unit 1 | Day 5 | TEK A2.2.A

Objective: I can graph and analyze the cubic and cube root parent functions, prove algebraically that each is an odd function, and describe how their rates of change differ.

Vocabulary

Odd function — satisfies __________________ for every x. Its graph has __________________ symmetry about the origin.

Even function — satisfies __________________. Its graph is symmetric across the ____________.

Average rate of change on [a, b] = ___________________________

Part 1 — Two Tables, One Relationship

x	-8	-2	-1	0	1	2	8
$f(x) = x^3$							

x	-8	-2	-1	0	1	2	8
$g(x) = \sqrt[3]{x}$							

Find a pair of entries showing the two functions undo each other: _____________________________

Part 2 — Prove Both Are Odd

Substitute -x into each rule and simplify. Do not just check one number.

a. $f(x) = x^3$. Find $f(-x)$:

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$f(-x)$ = _______________________________ so f is __________

b. $g(x) = \sqrt[3]{x}$. Find $g(-x)$:

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____

$g(-x)$ = _______________________________ so g is __________

c. Test $h(x) = x^2$ the same way. Is it odd, even, or neither? _____________________________

d. What does "odd" tell you about the graph without graphing it?

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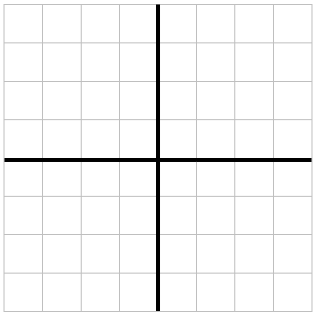
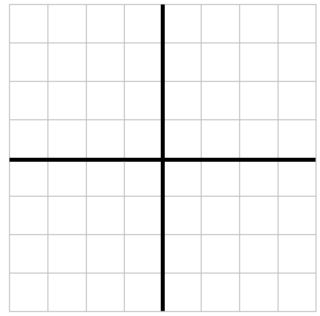
Part 3 — Graph Them Both

Both grids run -8 to 8 on each axis, so 1 square = 2 units.

A. $f(x) = x^3$



B. $g(x) = \sqrt[3]{x}$



Domain _____________ Range _____________

Mirror line between them: _____________

Part 4 — Rates of Change

Compute the average rate of change of $f(x) = x^3$ on each interval.

On [0, 1]: _______________ On [1, 2]: _______________ On [2, 3]: _______________

Now $g(x) = \sqrt[3]{x}$ on [0, 1]: _______________ On [1, 8]: _________________

Describe the contrast in one sentence:

____

The cubic _____________ as x grows while the cube root _____________ — which is what makes them mirror images.

Part 5 — Going Further

1. Solve $x^3 = 125$ and $x^2 = 25$. Count the real solutions for each.

$x^3 = 125 \rightarrow$ __________________________ $x^2 = 25 \rightarrow$ __________________________

2. Explain the difference using odd and even.

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x^3 is odd and never repeats an output; x^2 is even, so ______________________________

3. Which of the four parent functions so far accept negative inputs? ________________________

4. Can a function be both even and odd? Find one or prove none exists.

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Yes — ______________________________

Exit Ticket

A function has domain and range all reals, passes through the origin, and satisfies $f(-x) = -f(x)$.

Name two parent functions this could be: __________________________

Prove your choice is odd, algebraically: _______________________________________